年份	奖项	获奖者	分类方向	获奖理由	科学贡献
1918	物理	Max Planck	量子力学的建立	发现能量量子化	提出能量量子化假设，奠定了量子论的基石
1921	物理	Albert Einstein	量子力学的建立	发现光电效应定律	确立波粒二象性，证明光子的粒子属性
1922	物理	Niels Bohr	量子力学的建立	对原子结构及辐射的研究	提出玻尔模型，确立原子定态与能级跃迁
1929	物理	Louis de Broglie	量子力学的建立	发现电子的波动性	提出物质波理论，揭示微观粒子的波粒二象性
1932	物理	Heisenberg	量子力学的建立	创立量子力学	提出不确定性原理，建立矩阵力学体系
1933	物理	Schrödinger & Dirac	量子力学的建立	发现原子理论新形式	建立波动力学方程与相对论性量子力学
1954	物理	Max Born	量子力学的建立	对波函数的统计解释	确定了概率幅概念，解释了量子态本质
1922	物理	Niels Bohr	原子分子结构	对原子结构及辐射的研究	提出电子轨道量子化模型
1945	物理	Wolfgang Pauli	原子分子结构	发现不相容原理	解释了原子中电子壳层结构与元素周期律

年份	奖项	获奖者	分类方向	获奖理由	科学贡献
1954	化学	Linus Pauling	原子分子结构	化学键性质的研究	将量子力学引入化学，提出杂化轨道理论
1966	化学	Robert Mulliken	原子分子结构	建立分子轨道理论	发展了描述化学键的 LCAO-MO 方法
1998	化学	Kohn & Pople	原子分子结构	密度泛函理论	使大规模量子化学计算在材料建模中成为可能
2023	物理	Agostini & L’Huillier 等	原子分子结构	阿秒光脉冲	在极短时间尺度实时观测电子量子跃迁
1956	物理	Shockley & Bardeen 等	凝聚态物理	半导体与晶体管	基于能带理论控制电子量子态的里程碑
1972	物理	Bardeen & Cooper & Schrieffe	凝聚态物理	建立超导 BCS 理论	阐明了费米子配对产生的宏观相干态
1985	物理	Klaus von Klitzing	凝聚态物理	整数量子霍尔效应	揭示二维电子气在强磁场下的电阻量子化
1998	物理	Laughlin & Störmer & Tsui	凝聚态物理	分数量子霍尔效应	揭示了强关联系统中电子的复合准粒子行为
2016	物理	Thouless & Kosterlitz 等	凝聚态物理	拓扑相变理论	利用拓扑量子数描述物质的新奇物态

年份	奖项	获奖者	分类方向	获奖理由	科学贡献
1965	物理	Feynman & Schwinger 等	量子场论	量子电动力学	解决了发散问题，发明费曼图计算工具
1999	物理	’t Hooft & Veltman	量子场论	电弱相互作用结构	证明了非阿贝尔规范场论的可重整化性
2005	物理	Roy J. Glauber	量子信息	光学相干量子理论	定义了量子相干态，奠定量子通信基础
2012	物理	Haroche & Wineland	量子信息	操纵单量子系统	实现对单个原子或光子的非破坏性控制
2022	物理	Aspect & Clauser & Zeilinger	量子信息	纠缠光子与贝尔不等式	确证量子非局域性，开启量子纠缠工程应用
2025	物理	Immanuel Bloch	量子信息	超冷原子量子模拟	利用原子晶格复现复杂量子多体物理行为

The Paradigm of Quantum Field Theory: From Foundational Crisis to the Standard Model and Beyond

Yi Tang

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Abstract

Quantum Field Theory (QFT) represents the pinnacle of 20th-century theoretical physics, providing a unified language for particle physics and condensed matter systems. This review explores the transition from particle mechanics to field dynamics, the resolution of ultraviolet divergences through renormalization, and the current challenges posed by the hierarchy problem and the integration of gravity. We argue that QFT is not merely a set of calculations but a fundamental shift in our understanding of the vacuum and locality.

1 Introduction: The Ontological Shift

The core scientific problem of the early 20th century was the "causality crisis" created by merging Quantum Mechanics (QM) with Special Relativity. In non-relativistic QM, the position operator $\hat{x}$ is an observable, while time t is a parameter. This asymmetry violates the Lorentz invariance required by relativity.

Furthermore, the uncertainty principle $\Delta E \Delta t \geq \hbar/2$ implies that over short time intervals, energy fluctuations can exceed the rest mass of particles, leading to the creation of particle-antiparticle pairs. Thus, a single-particle description is mathematically inconsistent. QFT resolves this by promoting both space and time to parameters and treating the **field** $\phi(x,t)$ as the fundamental operator.

2 The Mathematical Framework: Path Integrals and Symmetry

The modern approach to QFT is largely governed by the **Feynman Path Integral** formulation. The transition amplitude is calculated by summing over all possible field configurations:

$$Z = \int \mathcal{D}\phi \, e^{i \int d^4x \mathcal{L}(\phi, \partial_\mu \phi)}$$

where $\mathcal{L}$ is the Lagrangian density. This framework highlights the role of **Symmetry**:

- **Global Symmetries:** Lead to conservation laws (Noether's Theorem), such as the conservation of charge.

- **Local (Gauge) Symmetries:** Dictate the existence of force-carrying bosons. For example, requiring local $U(1)$ invariance in the Dirac Lagrangian naturally necessitates the introduction of the photon field A_μ.

3 Breakthroughs: The Mastery of Divergences

A critical section of QFT history is the battle against "Infinities." When calculating self-energies, integrals often diverge at high momenta ($k \to \infty$), known as Ultraviolet (UV) divergences.

3.1 Feynman Diagrams: Visualizing Interaction

Prior to Richard Feynman, calculations in QFT were incredibly cumbersome. The introduction of Feynman diagrams allowed physicists to visualize complex particle interactions as topological graphs. Each line represents a propagator, and each vertex represents an interaction strength (coupling constant). This breakthrough turned abstract field equations into a systematic perturbative tool.

[Image of Feynman diagrams for electron-positron scattering]

3.2 Renormalization and the RG Flow

The breakthrough was realizing that "bare" parameters (mass and charge) are not observable. Through renormalization, we hide these infinities within the physical parameters. In the 1970s, Kenneth Wilson revolutionized this by introducing the **Renormalization Group (RG)**. He showed that QFTs are "scale-dependent." A theory might look like a collection of particles at one energy scale and a smooth fluid at another.

3.3 The Non-Abelian Revolution

Before the 1960s, QFT was largely limited to Electrodynamics. The work of Yang and Mills extended gauge theory to non-Abelian groups like $SU(2)$ and $SU(3)$. This allowed for the description of the "Strong" force, characterized by **Asymptotic Freedom**—where quarks become weaker as they get closer.

4 Current Achievements: The Standard Model

The representative achievement of QFT is the **Standard Model (SM)**, which describes the universe using the symmetry group:

$$SU(3)_C \times SU(2)_L \times U(1)_Y$$

[Image of the Standard Model of elementary particles chart]

This model has survived decades of scrutiny. The discovery of the Higgs Boson at the Large Hadron Collider (LHC) in 2012 confirmed the mechanism of **Spontaneous Symmetry Breaking**, proving that the vacuum itself possesses a structure that gives mass to elementary particles.

5 Future Outlook: Beyond the Effective Framework

Despite its success, most physicists now view QFT as an **Effective Field Theory (EFT)**. This means the SM is likely a "low-energy" approximation of a deeper theory.

5.1 Key Challenges

1. **The Hierarchy Problem:** Why is the Higgs mass (125 GeV) so much smaller than the Planck mass (10^{18} GeV)?
2. **Dark Matter:** Many hypothesize that Dark Matter consists of weakly interacting particles belonging to a "hidden sector" field not yet included in the Standard Model.
3. **Quantum Gravity:** Standard QFT fails when applied to Einsteins General Relativity because gravity is "non-renormalizable."

6 Conclusion

Quantum Field Theory remains the most accurate description of the subatomic world. Its future lies in the synthesis with gravity and the potential discovery of "Supersymmetry" or "Extra Dimensions" at higher energy scales. As an academic field, it remains the bridge between the infinitesimally small and the vast structure of the cosmos.